

Safety Sense: Enhancing Urban Cyclists' Perceived Safety during Motorcycle Overtaking through Augmented Reality Warning System

Yu-Ti Huang

Trans-disciplinary Bachelor Degree Program

National Taiwan University

Taipei, Taiwan

kvoyuti.93@gmail.com

Abstract

As cities embrace diverse mobility solutions, urban cyclists face significant safety challenges, particularly when interacting with speed-differential vehicles such as motorcycles in shared urban spaces. While Augmented Reality (AR) technology shows promise in improving cycling safety, its effectiveness in supporting urban cyclists during motorcycle overtaking events remains largely unexplored. This paper presents *Safety Sense*, an investigation of two AR warning modality combinations designed to enhance urban cyclists' perceived safety during motorcycle overtaking events: (1) a centralized visual display with warning sounds and (2) a bilateral visual display with voice prompts. Through a real-road study with 20 participants examining motorcycle overtaking events, this research evaluates the effectiveness of these combinations across three key dimensions: risk perception, directional awareness, and warning detection. Results show that both combinations significantly improved perceived safety compared to baseline conditions ($p < .001$, $\eta^2 > .69$), with the bilateral display and voice prompts outperforming the others in directional awareness and warning detection. Interview findings suggest that visual cues may be more effective than audio in facilitating rapid perception and response in complex urban traffic situations. This research offers valuable insights into multimodal AR warning design, contributing to the development of future AR-based safety technologies for vulnerable road users in evolving urban mobility contexts.

CCS Concepts

• Human-centered computing → Mixed / augmented reality.

Keywords

Augmented Reality, Urban Cycling Safety, Multimodal feedback

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1 Introduction

As cities embrace diverse mobility solutions, urban cyclists increasingly share roads with vehicles operating at different speeds. Data from the National Highway Traffic Safety Administration reveals that 85% of cyclist fatalities occur in complex urban environments [16]. While traditional traffic safety research focused on cyclist-car interactions, the adoption of micromobility solutions has created more complex speed-differential scenarios in shared spaces. The global shift toward sustainable transportation has led to a rise in both urban cycling and other forms of micromobility [9], creating situations where vulnerable road users of varying speeds must safely interact in shared spaces [11]. These speed-differential interactions, particularly overtaking events, not only pose significant safety threats [1] but also impact cyclists' perceived safety. Studies indicate that overtaking encounters are major contributors to cyclists' feelings of insecurity, which in turn affect their behavior, route choices, and willingness to ride [3, 25].

A significant safety concern for cyclists is their limited ability to detect and monitor approaching speed-differential vehicles in their surroundings [4]. This perceptual challenge is especially critical in motorcycle overtaking scenarios due to frequent interactions in mixed traffic lanes. Recent advancements in AR technology, particularly in AR glasses, present promising opportunities to address these challenges [8]. By overlaying information directly in cyclists' field of view, AR can deliver warnings without requiring riders to shift their attention from the road [2]. Designing effective AR warning interfaces for such scenarios is especially crucial [22], as cyclists must respond rapidly while maintaining awareness of the complicated traffic environment. However, there is still limited research in HCI on cyclist interactions with speed-differential vehicles, and existing studies on AR interfaces primarily focus on cyclist-car interactions or controlled simulator environments.

This paper presents *Safety Sense*, an investigation of two distinct AR warning systems designed to enhance cyclists' perceived safety during motorcycle overtaking events: (1) a consolidated visual display with warning sounds following established human factors design guidelines for vehicle interfaces [5], which emphasizes optimal warning placement to avoid distracting from critical visual information, and (2) a bilateral visual display with voice prompts that leverages AR's capability to present spatially-mapped warnings. The bilateral design builds on the human factors principle that spatial information is particularly suited for visual warnings [6], allowing cyclists to intuitively perceive the direction of approaching threats. Through a comparative evaluation with 20 participants in

real-road experiments, the research examines how these different approaches influence cyclists' perceived safety across three dimensions: risk perception, directional awareness, and warning detection. The findings contribute design insights for AR-based safety systems and advance understanding of how AR spatial information presentation and multimodal feedback can enhance perceived safety in diverse urban mobility contexts.

2 Related Work

2.1 Perceived Safety in Overtaking Events

Overtaking events are common but critical interactions that can result in severe and fatal consequences due to high impact speeds [17]. During overtaking scenarios, cyclists face various crash risks, particularly rear-ending and sideswiping incidents [18]. The passing phase is especially critical, as the overtaking vehicle gets closer to the cyclist, where inadequate lateral clearance can potentially destabilize the cyclist and create a perception of danger [20]. Through a field-test experiment, Llorca et al. demonstrated that cyclists' perceived risk during overtaking events is significantly influenced by both lateral clearance and overtaking speed [13]. However, current research on cyclists' perceived safety during overtaking events primarily focuses on car-bicycle interactions, leaving a significant gap in understanding motorcycle-bicycle interactions.

2.2 AR Solutions for Cyclist Safety

Prior HCI research has explored various AR technological interventions aimed at enhancing cyclists' perceived safety, demonstrating AR's potential benefits. For uncontrolled intersections, studies have explored visualization approaches, including X-ray vision to highlight occluded cars and countdown displays for crossing decisions [15], showing how AR can enhance cyclists' perceived safety in complex traffic situations. Studies of vehicle door-opening scenarios revealed that combining visual and audio feedback outperformed visual-only solutions [23], demonstrating their effectiveness for enhancing cyclist perceived safety. Research on child cyclists indicated that auditory cues were most easily understood with the fewest errors [14]. While AR's effectiveness has been shown in indoor environments and bicycle simulators, limited research exists on how multimodal approaches affect cyclists in real-world settings. Existing studies primarily focused on other age groups or conducted using bicycle simulators, leaving a gap in understanding AR applications for speed-differential overtaking scenarios. The research addresses these gaps by investigating two distinct AR warning system designs through real-road experiments using AR glasses. The study examines:

- **RQ1:** How do different AR warning display approaches (consolidated vs. bilateral) affect urban cyclists' perceived safety during motorcycle overtaking events?
- **RQ2:** What are the effects of multimodal feedback combinations (warning sounds vs. voice prompts) on perceived safety during motorcycle overtaking events?
- **RQ3:** How do these different AR warning approaches influence cyclists' risk perception, directional awareness, and warning detection during motorcycle overtaking events?

3 User Interview

3.1 Method

Semi-structured interviews were conducted with 16 regular urban cyclists (8 male, 8 female, ages 20-40, $M = 23.75$, $SD = 7.01$) to understand their safety challenges during motorcycle overtaking events. Participants were recruited through purposive and snow-ball sampling via local cycling groups, with the requirement of at least one year of urban cycling experience. This criterion ensured participants had sufficient exposure to urban traffic dynamics while still experiencing perceived safety concerns. The one-hour interviews followed a structured protocol exploring daily urban cycling challenges, perceived safety concerns during motorcycle interactions, and information needs during overtaking events. While none of the participants had prior AR glasses experience, 6 had used head-mounted displays, providing valuable perspective on wearable technology adoption. The interview data was analyzed using Braun and Clarke's thematic analysis framework [7].

3.2 Findings

The analysis revealed three interconnected factors influencing urban cyclists' perceived safety during motorcycle overtaking events. First, participants consistently emphasized concerns about speed differentials and close proximity. The close proximity creates pressure and heightens vulnerability, especially in congested traffic. As one participant noted, *"The way motorcycles overtake me with barely any space makes me feel fragile and unprotected"* (P3). This pressure was particularly acute regarding speed differences, with participants stating *"The significant speed difference between motorcycles and bicycles creates constant anxiety, especially when they pass by so quickly"* (P8). The lack of reaction time intensified this anxiety, as expressed by another participant, *"There's almost no time to react when motorcycles pass by, making it very stressful"* (P11).

Directional awareness emerged as a second critical challenge, particularly in perceiving approaching motorcycles. Participants described persistent uncertainty due to limited reaction time, *"In urban traffic, I can sense approaching vehicles but cannot immediately pinpoint their location, which creates constant uncertainty for me"* (P7). This challenge was compounded by motorcycles' movement patterns, as another participant noted, *"Motorcycles weave through traffic lanes, making it hard to predict their movements"* (P9). The combination of motorcycles' physical characteristics further complicated perception. A participant explained, *"With their small size and high speed, motorcycles often appear suddenly, making them hard to notice in time"* (P13).

The third factor centered on visual attention demands, with participants describing challenges in managing attention while cycling. One participant expressed the difficulty of maintaining both forward and rear awareness, *"I get tense every time I look back to check for vehicles. When I'm focusing on what's behind me, I might miss what's happening ahead"* (P4). The urban environment's complexity further challenged cyclists, as noted by one participant, *"There are too many things to watch out for, vehicles, motorcycles, pedestrians, road conditions, all while trying to ride safely"* (P12).

Regarding safety support requirements, participants highlighted the need for immediate, intuitive feedback that integrates with natural cycling behavior. Participants emphasized the importance

of receiving safety information without diverting attention from their primary cycling task. These insights directly informed the AR system design, leading to a focus on timely hazard alerts, clear directional indicators, and information presentation within cyclists' natural field of view.

4 System Design

4.1 Hardware

The detection system uses the HC-SR04 ultrasonic sensor, mounted on the bicycle seat post based on empirical studies of vehicle overtaking [22]. This mounting position was specifically chosen following the empirical studies, which demonstrated it as an optimal location for detecting approaching vehicles during overtaking events. The sensor has a detection range of 2-400 cm with a 15-degree field of view, as shown in the 15-degree sector in Figure 1. It detects motorcycles through ultrasonic pulse emission and echo timing. The system calculates the distance to approaching vehicles using:

$$distance = \frac{high - level\ time \times 340}{2}$$

The high-level time represents the pulse's round-trip duration, with sound speed approximately 340 m/s in air. The ESP32 microcontroller triggers a warning when a motorcycle enters the detection zone highlighted as the white sector during the overtaking event. As illustrated in Figure 4, the controlled experimental setup, where bicycles followed the white line while motorcycles moved from the white to yellow line during overtaking, ensured reliable detection without false alarms. When detecting significant distance changes within this zone, the Arduino immediately transmits the signal to the AR system via Wi-Fi, enabling real-time warnings through Unity-based visual and audio alerts.

4.2 AR Warning System

This research uses the XREAL Air2 Ultra AR glasses [26], chosen for their lightweight design (83g), making them ideal for real-world experiments. The AR warning system, comprising a mobile device and the AR glasses, was developed using the Unity engine and the NRS DK framework, provided by XREAL for application development on the glasses.

4.2.1 System 1. System 1, shown in Figure 2, adopts a consolidated warning approach featuring a central bicycle icon with directional indicators. The design follows established vehicle interface guidelines for safety warnings [5]. The visual component employs a centralized display strategy, which research has shown reduces cognitive load by presenting critical information in a single focal point [21]. The four-corner positioning options leverage peripheral vision capabilities [27], enabling cyclists to detect warnings without shifting visual attention from their primary riding task, a key principle in driver-vehicle interface design for minimizing distraction [12]. For the auditory component, System 1 implements simple warning tones, which align with multimodal warning research indicating that for lateral warning systems, visual-only modality may be insufficient [5]. These warning sounds were designed following guidelines that recommend non-speech audio signals for time-critical safety alerts where immediate reaction is required [10, 19]. This approach is consistent with research showing that drivers

generally do not find nuisance alarms annoying when presented via unobtrusive visual displays supported by appropriate auditory signals [5]. Considering individual differences in visual attention patterns and riding postures, the system enables customizable positioning of these visual warnings in the four corners of the cyclist's field of view. This customization addresses the design principle that safety systems accommodate diverse user preferences while maintaining effectiveness.

4.2.2 System 2. System 2, shown in Figure 3, was developed based on insights from preliminary user interviews and leveraging AR-specific capabilities. It features bilateral warning indicators with customizable positioning options at the top, middle, and bottom of the display, following research indicating that spatially mapped information can reduce cognitive load [27]. This approach directly addresses directional awareness challenges identified in the interviews, where participants described persistent uncertainty about approaching motorcycles' locations and movement patterns. The left and right indicators help users quickly identify the direction of an approaching motorcycle, creating intuitive spatial correspondence between the warning location and the hazard direction [6]. For audio feedback, the system incorporates voice prompts that clearly convey direction, such as "left" or "right," which follows guidelines that recommend using distinct directional auditory signals to promote appropriate orientation toward hazards [5]. This approach responds to participants' concerns about managing visual attention between forward and rear awareness while cycling, allowing them to maintain focus on the road ahead while receiving position information about approaching vehicles.

Both systems incorporate fluorescent yellow-green warning signals with flashing alerts to attract attention [21]. The selected yellow-green color maximizes attention-capturing properties while maintaining good contrast under various lighting conditions, and the flash frequency of 2-3 Hz is optimized for perceptual comfort while remaining highly noticeable [24]. The user-centered AR interface ensures consistent visibility of alerts regardless of head movement, addressing participants' need for immediate, intuitive feedback that integrates with natural cycling behavior without diverting attention from their primary cycling task.

5 Real-road Experiment

The study involved 20 regular urban cyclists (12 male, 8 female, ages 20-45, $M = 30$, $SD = 7.6$) who cycled in urban environments 4-6 days per week and had at least one year of cycling experience. All participants had normal vision and no prior experience with AR glasses. A within-subjects experiment compared baseline, system 1, and system 2 conditions on a 100-meter straight urban road segment with mixed-traffic lane markings. After providing informed consent and completing demographic questionnaires, participants familiarized themselves with the AR system. During the initial setup, participants could adjust the AR warning positions and size through the smartphone interface, allowing them to personalize the display based on their visual preferences. The final system maintained a fixed warning size for consistency.

Participants wore AR glasses connected to a pocket-secured smartphone and completed three randomized trials across different

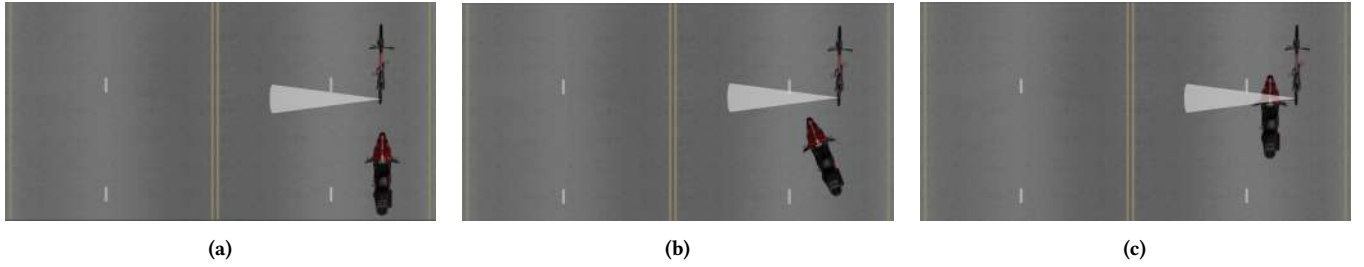


Figure 1: Sequential frames depicting the motorcycle overtaking scenario, with the ultrasonic sensor detection zone highlighted

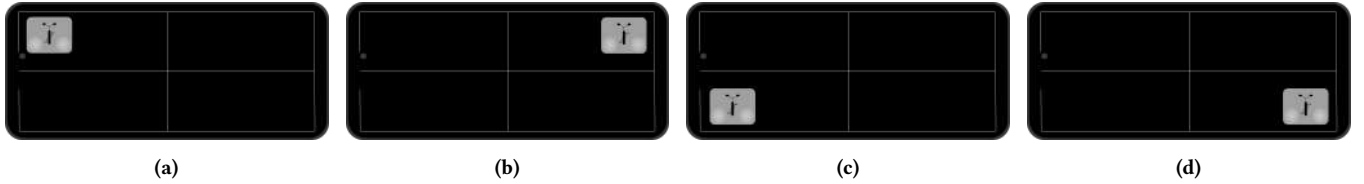


Figure 2: System 1's interface showing four selectable warning positions in the corners for AR display configuration

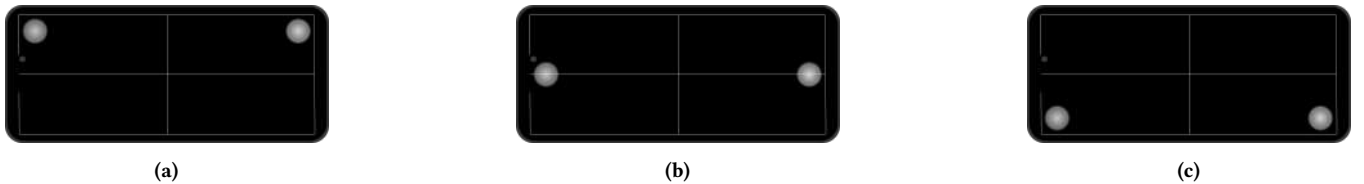


Figure 3: System 2's interface showing three selectable bilateral warning positions for AR display configuration

conditions. Each trial involved participants cycling along a designated route while a research assistant performed a controlled motorcycle overtaking at 30 km/h from the left side. Participants cycled along the yellow line, while the motorcycle passed along the white line to standardize lateral distance, as shown in Figure 4. To maintain ecological validity, the overtaking timing was not disclosed to participants, with system warnings activating automatically upon motorcycle detection. Post-trial feedback was collected through questionnaires using 7-point Likert scales and concluding semi-structured interviews.

5.1 Data Collection

A 7-point Likert scale questionnaire was developed to assess the effectiveness of the AR warning systems in enhancing urban cyclists' perceived safety during motorcycle overtaking events. The questionnaire measures three key dimensions derived from a preliminary study: (1) Risk Perception (e.g., "I can perceive potential risks when motorcycles overtake me"), (2) Directional Awareness (e.g., "I can identify the direction from which motorcycles are approaching"), and (3) Warning Detection (e.g., "I can notice the warning when the system activates").

To analyze the differences between baseline, System 1, and System 2 conditions, non-parametric statistical methods suitable for within-subject designs with ordinal data were employed. Specifically, the Friedman test was used to detect overall differences across

the three conditions. For pairwise post-hoc comparisons, Wilcoxon signed-rank tests with Bonferroni correction (significance threshold adjusted to $p < .017$) were conducted to control for Type I error rate inflation across multiple comparisons. This statistical approach ensured a robust and reliable assessment of the relative effectiveness of each system configuration.

5.2 User Study

5.2.1 Quantitative Results. The results suggest that both AR warning systems effectively enhanced perceived safety. Friedman tests revealed significant improvements across all three dimensions: Risk Perception ($\chi^2 = 28.4$), Directional Awareness ($\chi^2 = 25.9$), and Warning Detection ($\chi^2 = 24.8$), all with $p < .001$. These results indicate statistically significant differences between conditions, with large effect sizes (η^2 ranging from 0.69 to 0.79), suggesting substantial practical significance of these improvements. Post-hoc Wilcoxon tests with Bonferroni correction ($p < .017$) revealed that both systems significantly improved cyclists' perceived safety compared to the baseline condition. Notably, System 2's bilateral visual display with voice prompts demonstrated superior performance in Directional Awareness ($Mdn = 6.0$, $IQR = 1.0$ vs. $Mdn = 5.0$, $IQR = 2.0$) and Warning Detection ($Mdn = 6.0$, $IQR = 2.0$ vs. $Mdn = 5.0$, $IQR = 2.0$) compared to System 1's centralized visual display with warning sounds ($p < .05$). However, both systems achieved similar improvements in Risk Perception.

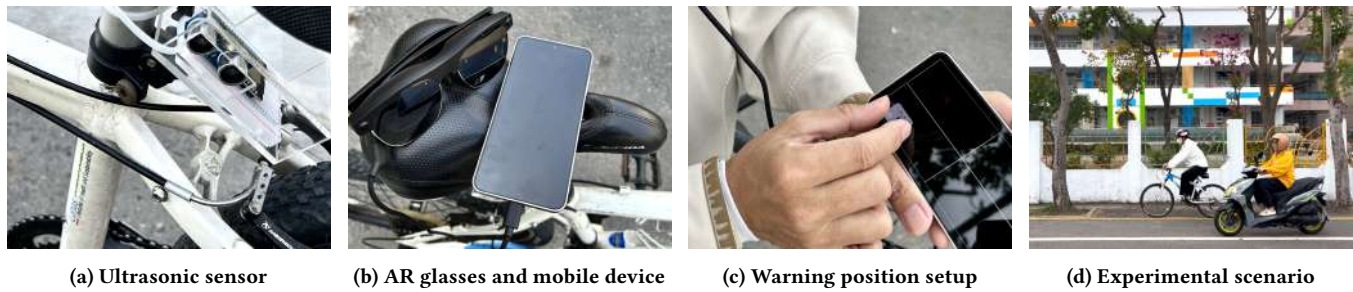


Figure 4: Real-road experimental setup showing a participant cycling with AR glasses of motorcycle overtaking



Figure 5: Cyclist's view with AR warning interface overlay

5.2.2 Qualitative Feedback. The user evaluation revealed clear preferences between the two AR warning designs. System 2 was preferred by 75% of participants (15/20), while System 1 was preferred by 25% (5/20). Most of the participants favored System 2, primarily

because of the visual display, rather than the audio. They considered the visual component as the primary source of information, with audio serving as secondary support during overtaking scenarios. As P16 explained, “While focusing on the front, the visual change in

Table 1: Friedman and Wilcoxon Test Results of Perceived Safety Questionnaire (N=20)

Dimensions	Baseline Mdn (IQR)	System1 Mdn (IQR)	System2 Mdn (IQR)	χ^2	p	η^2
Risk Perception	2.0 (1.0)	5.5 (2.0)	5.5 (2.0)	28.4	<.001**	.79
Directional Awareness	2.5 (2.0)	5.0 (2.0)	6.0 (1.0)	25.9	<.001**	.72
Warning Detection	3.0 (2.0)	5.0 (2.0)	6.0 (2.0)	24.8	<.001**	.69

^a $p < .05$; ** $p < .01$; Mdn = Median; IQR = Interquartile Range; Effect size interpretation: η^2 (> .06: medium, > .14: large); B = Baseline, S1 = Centralized visual display with warnings sounds, S2 = Bilateral visual display with voice prompts; post-hoc comparisons were performed using Wilcoxon signed-rank tests with Bonferroni correction ($p < .017$)

my view catches my attention. But the difference lies in how the visual cues are presented, as it helps me recognize the meaning without other thinking". Most participants viewed the audio component as a reminder, signaling the presence of an alert. However, there was no specific preference for voice prompts versus warning sounds. Notably, participants who preferred System 2 mentioned that the centralized visual display did not significantly affect them, while the audio in System 1 helped them notice the alert.

For System 2, which features a bilateral visual display with voice prompts, 75% of the participants strongly preferred it. The preference was mainly due to its intuitive spatial mapping and clear directional information. As P4 stated, *"The distributed warnings immediately allow me to intuitively recognize the direction"*. For System 1, which employs a centralized visual display with warning sounds, 25% of the participants preferred it. These participants appreciated the focused attention point and the simplicity it provided, with fewer distractions. P12 explained, *"Having warnings in a single location feels less overwhelming and keeps my view clean without affecting forward vision or causing blind spots"*. However, participants who preferred System 2 noted that this required additional processing time and delayed their reaction. As P4 mentioned, *"I cannot immediately identify the direction when it lights up. When the display is in the left corner, I tend to subconsciously think it is indicating the left, and I need to pay attention to confirm whether it's actually the left or right"*. Regarding the positioning of the AR warnings, participants exhibited diverse preferences in their configuration choices, with no consistently preferred location. This suggests that individual preferences varied, and a one-size-fits-all approach might not be ideal. In addition, most of the participants rarely ever adjusted the warning size, indicating that the default scale was generally suitable and did not require frequent customization.

6 Discussion

The findings from this study provide valuable insights into the role of both visual and audio elements in enhancing urban cyclists' perceived safety. Regarding risk perception, both systems showed similar improvements, indicating the combination of visual and audio alerts effectively enhances cyclists' perception of risk. However, System 2 outperformed System 1 in specific dimensions, including directional awareness and warning detection, highlighting the advantages of System 2's design. This section discusses the contributions of visual and audio cues and their design implications for future AR systems. From a visual design perspective, System

2's bilateral display effectively leverages the spatial advantages of AR technology. By positioning the warnings in alignment with the real-world environment, users can quickly interpret the information with minimal cognitive effort. This spatial mapping helps users intuitively grasp the relative location of hazards, as opposed to System 1's centralized display, which requires more focused attention. The centralized display was less effective in situations where rapid comprehension and response were crucial, as users had to focus specifically on the display, potentially increasing cognitive load and delaying reaction time. For audio design, System 2's directional voice prompts and System 1's warning sounds showed minimal difference in supporting cyclists. Interview findings revealed that visual warnings played a more critical role in emergency situations, with audio feedback serving as a supplementary aid. This finding is particularly important in urban cycling contexts, where cyclists need to focus on complex traffic conditions and typically keep their attention forward.

7 Conclusion

The findings show that both AR systems significantly improved perceived safety across all dimensions, with the combination of bilateral display and voice prompts particularly excelling in supporting directional awareness and warning detection. The study highlights the potential of AR's spatial mapping capabilities, with visual cues proving more impactful than audio feedback in these specific scenarios. While these results are promising, there are some limitations, including the relatively small sample size (N=20) and the focus on perceived safety rather than actual safety outcomes. Despite these limitations, the findings offer valuable insights for the design of multimodal AR safety systems, contributing to the ongoing development of innovative technologies aimed at vulnerable road users in evolving urban mobility contexts. Future research could build upon these findings to explore real-world applications and further refine the integration of AR technologies in enhancing urban transportation safety.

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